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18.02 Multivariable Calculus
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18.02 Lecture 18. – Tue, Oct 23, 2007

Change of variables.

Example 1: area of ellipse with semiaxes a and b : setting $u = x/a$, $v = y/b$,

$$\iint_{(x/a)^2 + (y/b)^2 < 1} dx dy = \iint_{u^2 + v^2 < 1} ab du dv = ab \iint_{u^2 + v^2 < 1} du dv = \pi ab.$$

(substitution works here as in 1-variable calculus: $du = \frac{1}{a} dx$, $dv = \frac{1}{b} dy$, so $du dv = \frac{1}{ab} dx dy$).

In general, must find out the scale factor (ratio between $du dv$ and $dx dy$)?

Example 2: say we set $u = 3x - 2y$, $v = x + y$ to simplify either integrand or bounds of integration. What is the relation between $dA = dx dy$ and $dA' = du dv$? (area elements in xy - and uv -planes).

Answer: consider a small rectangle of area $\Delta A = \Delta x \Delta y$, it becomes in uv -coordinates a parallelogram of area $\Delta A'$. Here the answer is independent of which rectangle we take, so we can take e.g. the unit square in xy -coordinates.

In the uv -plane, $\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$, so this becomes a parallelogram with sides given by vectors $\langle 3, 1 \rangle$ and $\langle -2, 1 \rangle$ (picture drawn), and area = $\det = \begin{vmatrix} 3 & 1 \\ -2 & 1 \end{vmatrix} = 5$ $\left(= \begin{vmatrix} 3 & -2 \\ 1 & 1 \end{vmatrix} \right)$.

For any rectangle $\Delta A' = 5\Delta A$, in the limit $dA' = 5dA$, i.e. $du dv = 5dx dy$. So $\iint \dots dx dy = \iint \dots \frac{1}{5} du dv$.

General case: approximation formula $\Delta u \approx u_x \Delta x + u_y \Delta y$, $\Delta v \approx v_x \Delta x + v_y \Delta y$, i.e.

$$\begin{bmatrix} \Delta u \\ \Delta v \end{bmatrix} \approx \begin{bmatrix} u_x & u_y \\ v_x & v_y \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}.$$

A small xy -rectangle is approx. a parallelogram in uv -coords, but scale factor depends on x and y now. By the same argument as before, the scale factor is the determinant.

Definition: the Jacobian is $J = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$. Then $du dv = |J| dx dy$.

(absolute value because area is the absolute value of the determinant).

Example 1: polar coordinates $x = r \cos \theta$, $y = r \sin \theta$:

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} x_r & x_\theta \\ y_r & y_\theta \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta = r.$$

So $dx dy = r dr d\theta$, as seen before.

Example 2: compute $\int_0^1 \int_0^1 x^2 y dx dy$ by changing to $u = x$, $v = xy$ (usually motivation is to simplify either integrand or region; here neither happens, but we just illustrate the general method).

1) Area element: Jacobian is $\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ y & x \end{vmatrix} = x$, so $du dv = x dx dy$, i.e. $dx dy = \frac{1}{x} du dv$.

2) Express integrand in terms of u, v : $x^2 y dx dy = x^2 y \frac{1}{x} du dv = xy du dv = v du dv$.

3) Find bounds (picture drawn): if we integrate $du dv$, then first we keep $v = xy$ constant, slice looks like portion of hyperbola (picture shown), parametrized by $u = x$. The bounds are: at the top boundary $y = 1$, so $v/u = 1$, i.e. $u = v$; at the right boundary, $x = 1$, so $u = 1$. So the inner

integral is $\int_v^1$. The first slice is $v = 0$, the last is $v = 1$; so we get

$$\int_0^1 \int_v^1 v \, du \, dv.$$

Besides the picture in xy coordinates (a square sliced by hyperbolas), I also drew a picture in uv coordinates (a triangle), which some students may find is an easier way of getting the bounds for u and v .